

Reflections on Creating

From Launch to Litter: Orbital Overload



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Introduction

This project involved collecting, cleaning, and analysing satellite and space debris datasets to explore the risks posed by orbital congestion, and the creation of visualisations, and presentation design. The report and related presentation aim to describe the Kessler Syndrome, to explore what is in orbit, who put it there and how long it lasts.

I feel that I have met the learning objectives of this module, as this reflection report will attest.

I used standard data cleaning and analysis techniques to produce a working dataset. I used python in Jupyter notebooks to analyse and visualise this data. I created numerous charts and maps using a variety of visualisation libraries.

The charts were designed to display the data in an engaging way. The choice of type was led by the structure of the data being displayed, as well as the focus of the message being conveyed.

In this reflection on the project, I will evaluate the experience of creating the report and presentation. I will discuss the skills I have earned, and the learning that I have done.

Reflection

I believe that I chose a timely and relevant topic and created a presentation that explores it in an engaging and interesting manner.

Activities

Although I am interested in space exploration, technology and science, I was not initially familiar with all the terms being used in discussions around the subject. I spent some time early in the project timeframe reading to research domain knowledge.

It was more challenging than expected to find suitable data; I obtained and explored several datasets before finding an up to date and complete dataset. On one occasion, I opted to avoid scraping a complete dataset from a non-commercial website due to ethical concerns about the bandwidth and processing costs that they would incur.

If undertaking this project again, I would have started exploration of the available datasets earlier. Once I found a suitable dataset, work progressed quickly. I had assumed that NASA would provide suitable data, but this was not the case.

The dataset I finally used came with extensive documentation which I needed to use to interpret the codes used in various categorial data fields. I replaced these codes with descriptive text to aid understanding.

The data exploration, cleaning, feature engineering and visualisation was conducted on Linux and MacOS systems using VSCode and Jupyter notebooks. Charts were saved as PNG files. GitHub was used for multi-system version control. The final presentation was assembled on a MacBook using Microsoft PowerPoint. This was a flexible and productive way for me to work, allowing me to make the best use of the equipment I have available.

I used a combination of different chart types, including horizontal and vertical bar charts, bubble charts, doughnut charts and choropleth maps. I chose the chart type depending on the type of data and the number of categories being compared. I found coloured bubble charts helpful to show categorial data over time.

I learned some new techniques when working with fine detail in creating graphs in Matplotlib. For example, I used controls to influence the relative positioning of the chart data, text labels and axes. I learned how to use HTML-like coding to adjust text styles, and how to combine data from different series into text labels. This was the only time in the project that I used AI; I tend to use it like an 'extended Google' resource.

I split the data engineering and visualisation tasks into two Jupyter notebooks. I should have focussed on finishing the data cleaning and feature engineering steps earlier in the project timeline. Then I would have been able to focus more time on producing charts. I found myself going back to the data engineering notebook to make changes quite late into the project. I should also have created multiple smaller visualisation notebooks to save processing time.

Overall, I feel that I have developed as a data scientist through this project; it has been a valuable exercise. I found the class discussions useful, especially the workshops around time series and the analysis of geo-spatial data.

To summarise the above:

Strengths

- Ability to research a topic to gain domain knowledge.
- Persistence in overcoming dataset availability challenges.
- Technical skills in Python, pandas, visualisation libraries, and PowerPoint.
- Showed ethical awareness when I avoided scraping data excessively.
- Creativity in use of colour and chart design.

Weaknesses

- Time management (finding suitable datasets and then data cleaning took longer than expected).
- Initial lack of familiarity with satellite terminology and datasets.
- I use LLMs like a 'clever Google' rather than being an AI native, which would likely speed me up.

Skills Gained

- Improved at data cleaning and feature engineering.
- Mapping categorical codes to meaningful labels.
- Geocoding and handling API constraints.
- Improved python coding skills, such as more use of lambda.
- Learned how to apply colours to charts.
- Working with fine detail in positioning graph elements and creating custom category labels.

Presentation

I used PowerPoint to create my presentation storyboard. I have created many hundreds of presentations over the years, and I find the slide sorter is useful to organise ideas. Slides are easily rearranged, added and removed. From experience, presentations should aim follow a story.

Once defined, the structure and slide titles suggest the graphs and infographics that would best illustrate the point being made. The titles are positioned so to give the maximum space for the charts.

Colour is a key component of data visualisation. The colour scheme was chosen to be vibrant and engaging. Bright, high-contrast colours were chosen to stand out vividly against the dark blue background as shown below. The blue background is designed to evoke a 'space theme' without being as stark as pure black.



There are no interactive charts in the presentation. This is a deliberate choice from years of experience delivering training courses and technical presentations. I would find 'driving' an interactive chart fraught with danger during a live presentation.

For this purpose, I would prefer to pre-generate an animation showing data changing on a chart. For this presentation I created a simple animation within PowerPoint to show a cascade.

If we had been asked to produce a website or dashboard specifically, I would have made my charts interactive to allow the user to explore the data in their own way. However, in a presentation the story is led by the presenter, so there was no need.

Once the visuals were complete, the narration track was recorded. I chose not to put video of myself into the presentation as there would be no suitable place on most slides to accept the 'Cameo' video recording without overwriting a graph visual.

I wanted the presentation to have a 'space feel', so I experimented with several 'ambient' music tracks as backing music and was able to find some suitable royalty free music. I needed to use ffmpeg on my MacBook to reduce the music audio level and to create a mono version of the final video. The video file was uploaded to YouTube¹.

The final elements of the story were not told directly with charts, but through words drawn from research, backed up by facts from my analysis of the data.

¹ <https://youtu.be/OMqgP5OI7jo>

Way Forward

The projects that I have carried out so far in this degree program, including this project, have served to give me a strong basis from which to work with large, complex and messy datasets in the future. I have learned that I am able to adapt and persevere through new technical challenges.

I have learned that I can work well with python to produce interesting and visually appealing charts. When I started this degree, I had never used pandas to manipulate data before. I now realise that I have progressed a long way and gained some valuable skills. My python programming was acceptable before, but I am now happy to use more complex features such as lambda functions.

There are a variety of visualisation libraries that I have explored in detail in this project. I now know how to produce composite labels, to change fonts and chart colours, and to create custom gradient colour maps.

I have created a template Jupyter notebook that includes repeatable functions for all the techniques that I have learned.

I am more expert in using python for data analysis and visualisation than other tools I have available. I will aim to do some workplace projects in other tools, like Tableau, PowerBI, and the 'R' language to give myself a broader range of experience to draw from. I will also spend some time strengthening my statistical analysis skills.

I should also learn to use the AI tools that I have at my disposal to help me write code. This would speed up the data analysis stages and assist me to create graphs more quickly with fewer iterations.

This all applies to and complements my work in cyber security, as often our data is unstructured and complicated to process.